

DBMS Assignment-3

Section - A : SQL & Query concepts

- Q1 What is SQL? Why is it important?
- Ans
- SQL (Structured Query language) is a language used to communicate with database.
 - It helps to:
 - Create table
 - Insert data
 - Update/Delete data
 - Retrieve data
 - Used in real world system like banking, apps, websites
 - Makes data handling easy & efficient.

Q2 Difference b/w DDL & DML?

Ans DDL (Data definition language):

- Used to define database structure
- Commands : Create, alter, Drop

DML (Data Manipulation language):

- Used to manage data inside tables.
- Commands : Insert, Update, Delete

Q3 What is Join? Explain its types

Ans Join is used to combine data from two or more tables using a common column

Types:

- Inner JOIN → only matching rows
- Left JOIN → all rows from left table + matching from right
- Right JOIN → all rows from right table + matching from left
- Full JOIN → all rows from both tables.

Q4 What is Group BY and Having clause?

Group BY

- ↳ Groups rows with same values
- ↳ find total salary per department

HAVING:

- ↳ Filters grouped results
- ↳ filters only high totals.

Q5 What is Subquery?

Ans → A procedural language used in DBMS

→ Performs operations like:

- Select
- Project
- Join

Q7 Difference b/w WHERE & HAVING?

WHERE

- ↳ Filters rows before grouping
- ↳ individual rows

HAVING

- ↳ Filters after grouping
- ↳ grouped data.

Use Case

Section-B

1 Retrieve students with marks > 75

SELECT * FROM Students
WHERE marks > 75;

↓
for applying condition

2 JOIN Student & Course tables

SELECT s.name, c.course_name
FROM Student s
JOIN Course c
ON s.course_id = c.course_id;

3 Count students per department

SELECT dept-id, count(*)
FROM Students;
GROUP BY dept-id;

Section-C

1 Employee in a specific department

Code SELECT e.emp-name, d.dept-name
FROM Employee e
JOIN Department d
ON e.dept-no = d.dept-no
WHERE d.dept-name = 'HR';

2
Code Total sales per customer

```
SELECT customer-id, SUM(amaunt)
FROM orders
GROUP BY customer-id;
```

3 Top 3 students

```
SELECT * FROM students
ORDER BY marks DESC
LIMIT 3;
```

Q1 Write a SQL query for any real-life system and explain logic?

Ans Sec-D
SELECT customer-id, SUM(orders-amount) AS total-spend
FROM orders
GROUP BY customer-id;

Logic

- ↳ This query is used in a e-commerce system
- ↳ It calculates the total money spent by each customer

- SELECT customer-id → Select each customer
- SUM(orders-amount) → adds all orders of that customer
- GROUP BY customer-id → groups all orders belonging to same customer

Q2 Write JOIN query & mention one use case?

Ans Select e.emp-name, d.dept-name
FROM Employee e
JOIN Departments d
ON e.dept-no = d.dept-no;

Use Case

- ↳ Display employee details with department
- ↳ Generate reports
- ↳ Manage organizational data.

Q3 Write a GROUP BY query & explain its output.

Ans SELECT dept-no, COUNT(*) AS total_employees
FROM Employees
GROUP BY dept-no;

Output

dept-no	total_employees
10	3
20	2
30	25

Explanation

- ↳ all employees having the same department no. are grouped together
- ↳ Then COUNT(*) calculates the no. of emp in each grp.
- ↳ So instead of showing every emp, it shows a summary.

VIVA components

Q1 Why is SQL imp in real world applications?

Ans SQL is used to store, manage & retrieve data from database

- It is used in almost every system like → Banking systems
→ E-commerce websites
- It helps to → Insert, delete & update
↳ Maintain data accuracy & consistency.
- SQL supports — Security
— Data analysis

Real-life Eg

- ↳ Calculate total sales.
- ↳ Manage product data.

Q2 What happens if JOIN condition is incorrect?

A2 The query may produce wrong or meaningless result.
It may return extra rows.
It may ~~not~~ miss required records.

Q3 Give me real-life eg of GROUP BY?

A3 Group BY is used whenever we need summary data.
↳ Total sales per branch
↳ Avg. salary per department
↳ No. of students per class

Q4 What is the impact on performance of complex queries?

A4 → Complex queries can slow down database performance.
→ They require more processing time to execute.
→ Use more CPU and memory resources.

See-A DBMS CONCEPTS

Q1 What is DBMS & why required?

A1 → DBMS is software to store and manage data.
→ Required for
↳ Security
↳ Consistency
↳ Easy access.

Q2 DBMS vs File system

- ↳ Reduces duplication
- ↳ Provides security
- ↳ Offers backup ~~and~~ recovery.

Q3 Data Abstraction

↳ Hides complexity

↳ Levels:

↳ Physical → storage

↳ Logical → Structure

↳ View → user

Q4 Three-Schema Architecture

↳ Internal → storage

↳ Conceptual → Structure

↳ External → user view

Q5 Data Independence

↳ change one level without affecting others.

Q6 ER Model

↳ Represents:

↳ Entities

↳ Attributes

↳ Relationships

Q4 Integrity constraints

↳ Rules to maintain correct data

↳ Eg:

↳ Primary key

↳ Foreign key